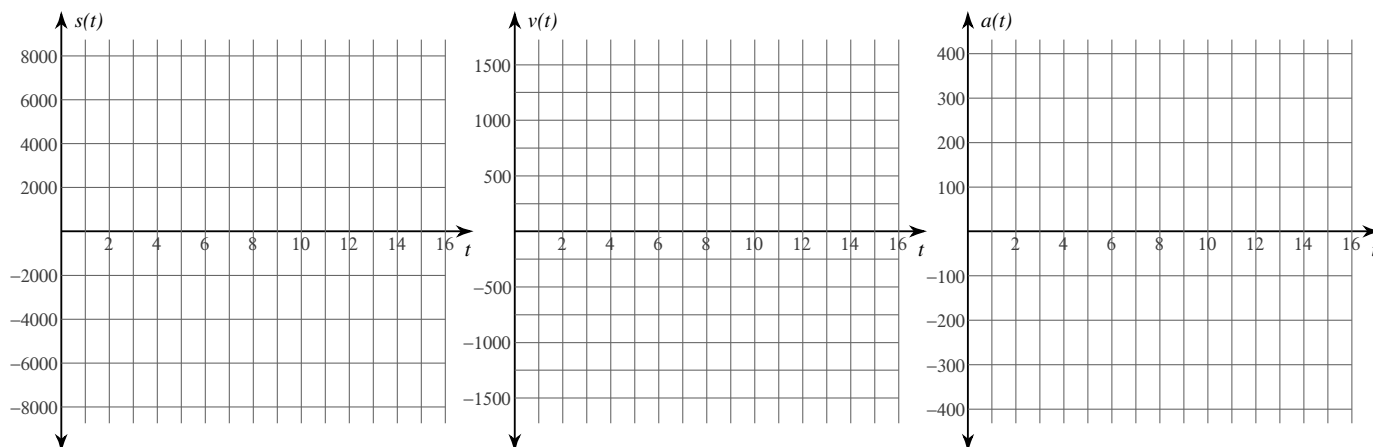


A particle moves along a coordinate line. Its acceleration function is $a(t)$ for $t \geq 0$. For each problem, find the position, velocity, speed, and acceleration at the given value for t . You may use the blank graphs to sketch $s(t)$, $v(t)$, and $a(t)$.

3) $a(t) = -12t^2 + 72t$; $s(0) = 0$; $v(0) = 0$; at $t = 5$



A particle moves along a coordinate line. Its acceleration function is $a(t)$ for $t \geq 0$. For each problem, find the displacement of the particle and the distance traveled by the particle over the given interval. You may use the blank graphs to sketch $v(t)$ and $a(t)$.

4) $a(t) = 6t - 48$; $v(0) = 144$; $1 \leq t \leq 8$

